Nmap Timing and Performance — Polite and Normal Scans

Aim

To execute and display Nmap timing and performance commands in Kali Linux for the polite and normal scan timing templates listed in the experiment.

Requirements

Kali Linux, Nmap, terminal access, and a target system that you own or have explicit permission to scan.

Safety Note

Use these commands only on your own computer, an authorized lab virtual machine, or a network for which you have permission. This experiment demonstrates scan timing; it does not guarantee evasion of intrusion detection systems (IDS).

Procedure

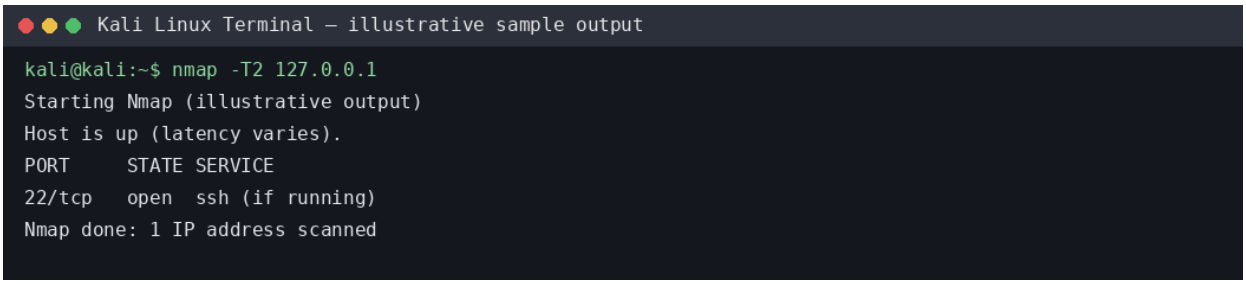
1. Open Kali Linux and launch the Terminal.
2. Check Nmap installation using: nmap --version
3. For the polite timing template, run: nmap -T2 127.0.0.1
4. Observe the scan output and note the open, closed, or filtered ports shown.
5. For the normal timing template, run: nmap -T3 127.0.0.1
6. Compare the commands and outputs. Results depend on the target host and its running services.

(i) Polite Timing Template

Command:

nmap -T2 127.0.0.1

Explanation: The -T2 timing template is called “polite.” It slows down scan activity to reduce network load and is generally slower than the normal template. It is a timing option, not a guarantee that an IDS will not detect the scan.



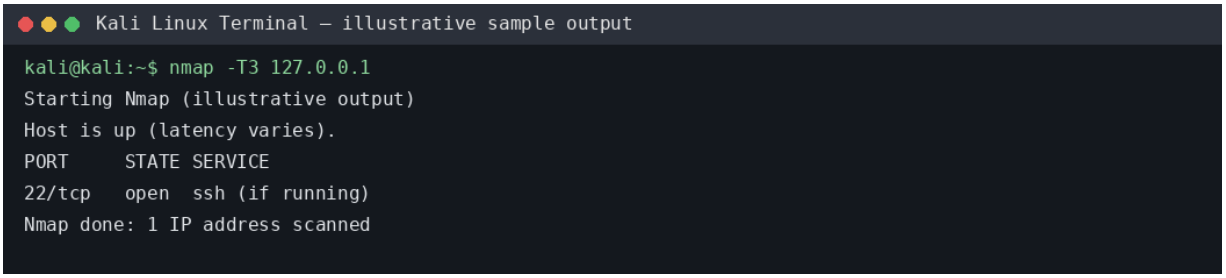
Illustrative sample output only. Replace with a screenshot from your own authorized lab scan.

(ii) Normal Timing Template

Command:

nmap -T3 127.0.0.1

Explanation: The -T3 timing template is Nmap's default normal timing profile. Nmap adjusts scan speed based on network conditions and host response.



Illustrative sample output only. Actual results will vary depending on the target and services.

Observation

The polite (-T2) timing template is slower and more conservative than the normal (-T3) template. Port results may be the same or different depending on the target and network conditions.

Result

The Nmap polite and normal timing commands were studied and executed as examples. Actual scan results should be recorded from the user's authorized Kali Linux lab environment.